
5(a)	The induced e.m.f. is directly proportional to the rate of change of magnetic flux linkage.	B1
5(b)	As the seismic waves passes, the vibrations moves the case (and coil) and causes the movement of the case (and coil) relative to the magnet.	B1
	This results in a change of magnetic flux linkage through the coil, and by Faraday's law, an e.m.f. will be induced across the coil.	B1
5(c)	$\varepsilon = \frac{dN\phi}{dt} \Rightarrow \frac{d\phi}{dt} = \frac{\varepsilon}{N}$ $= \frac{85 \times 10^{-3}}{50}$ $= 1.7 \times 10^{-3}$	C1 A1
5(d)	✓ magnet with stronger magnetic field ✓ coil with many turns ✓ spring with a lower spring constant. (any two of the above)	B1 B1

Question		Marks
6(a)	The r.m.s value of an a.c. will give the same heating effect on a resistor as	B1